

Weekly Assignment: Logic (Epp 2.1 & 2.2)

Complete each of the following problems. Show all work and justify your answers.

Problem 1 (Epp 2.1 #4). Represent the common form of each argument using letters to stand for component sentences, and fill in the blanks so that the argument in part (b) has the same logical form as the argument in part (a).

- a. If the program syntax is faulty, then the computer will generate an error message.
If the computer generates an error message, then the program will not run.
Therefore, if the program syntax is faulty, then the program will not run.
- b. If this simple graph _____, then it is complete.
If this graph _____, then any two of its vertices can be joined by a path.
Therefore, if this simple graph has 4 vertices and 6 edges, then _____.

Problem 2 (Epp 2.1 #8 b,c,e). Write the following statements in symbolic form using the symbols $\neg$, $\vee$, and $\wedge$. Let h = "John is healthy," w = "John is wealthy," and s = "John is wise."

- b. John is not wealthy but he is healthy and wise.
- c. John is neither healthy, wealthy, nor wise.
- e. John is wealthy, but he is not both healthy and wise.

Problem 3 (Epp 2.1 #9 b,c). Write the following statements in symbolic form using the symbols $\neg$, $\vee$, and $\wedge$. Let p = " $x > 5$," q = " $x = 5$," and r = " $10 > x$."

- b. $10 > x > 5$
- c. $10 > x \geq 5$

Problem 4 (Epp 2.1 #22). Determine whether the following statement forms are logically equivalent. Construct a truth table and include a sentence justifying your answer.

$$p \wedge (q \vee r) \quad \text{and} \quad (p \wedge q) \vee (p \wedge r)$$

Problem 5 (Epp 2.1 #24). Determine whether the following statement forms are logically equivalent. Construct a truth table and include a sentence justifying your answer.

$$(p \vee q) \vee (p \wedge r) \quad \text{and} \quad (p \vee q) \wedge r$$

Problem 6 (Epp 2.1 #42). Use a truth table to determine whether the following statement form is a tautology, a contradiction, or neither.

$$((\neg p \wedge q) \wedge (q \wedge r)) \wedge \neg q$$

Problem 7 (Epp 2.2 #2). Rewrite the following statement in if-then form:

I am on time for work if I catch the 8:05 bus.

Problem 8 (Epp 2.2 #18). Write each of the following three statements in symbolic form and determine which pairs are logically equivalent. Include truth tables and a few words of explanation.

- (i) If it walks like a duck and it talks like a duck, then it is a duck.
- (ii) Either it does not walk like a duck or it does not talk like a duck, or it is a duck.
- (iii) If it does not walk like a duck and it does not talk like a duck, then it is not a duck.